

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Indicative content • Refrigeration/chilling – slows down (but does not stop) bacterial growth. Do not accept: stops growth • freezing – stops bacteria multiplying (but does not kill them) Do not accept: slows growth • drying/smoking – removes water (so bacteria cannot digest and absorb the food source) • salting / curing– bacteria lose water (from their cells so they dehydrate and cannot reproduce) • pickling – (addition of vinegar to) lower pH (and inactivate microorganisms.) • Accept: irradiation of food – kills bacteria • Allow ref to removal of oxygen – stops/kills bacteria 5-6 marks 2/3 methods with 2/3 correct explanations. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Up to 3 methods with at least 1 correct explanation. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Up to 3 methods stated but no explanations. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						

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	(b)	(i)		2 800 (1) 4 (1)		2		2	2	2
		(ii)		UHT – no bacteria present (1) Nothing to grow / temperature doesn't need to be lowered (and storage time is long so agree) (1)			2	2		
				Question 4 total	6	2	2	10	2	2